

# BharatTrip Refund Escalations: Business Case & Solution Proposal

**Executive Summary (TL;DR)** BharatTrip is experiencing a spike in customer/agent refund escalations despite stable refund volumes. Our analysis identified a severe operational disconnect between Support and Finance, leading to **100 dropped tickets** and **149 short payments**. By implementing a **Single Source of Truth (SSOT)** and an **AI-Driven Data Reconciliation Workflow**, we can eliminate manual data leakage, automate agent communications, and project an 80% drop in escalations over the next quarter without adding headcount.

## 1. Defining the Problem

Customer and agent escalations regarding refunds are rising. The core underlying problem is a **severe operational disconnect between the Support and Finance teams**, characterized by siloed tracking systems and inconsistent definitions of refund statuses.

### Key Communication Failures:

- 1. **Data Leakage:** Informal requests via WhatsApp and Email bypass the tracker altogether, meaning they are never reviewed or processed.
- 2. **"Closed" Status Ambiguity:** The Support team marks tickets as "Closed" once handed off to Finance, misleading the agent into believing the refund has been disbursed.
- 3. **Hidden Policy Deductions:** Finance applies necessary deductions (e.g., cancellation fees) and processes payouts, but this information is never relayed back to the agent by Support, causing "short payment" escalations.

## 2. Data Analysis & Insights

By cross-referencing the Support\_Tracker, Finance\_Tracker, and Escalations logs via a deep database join, we identified the specific patterns driving the **155 total escalations**.

[image]

### The Discrepancy Breakdown

| Escalation Driver    | Ticket Volume | Description                                                          |
|----------------------|---------------|----------------------------------------------------------------------|
| Dropped Handoffs     | 100           | Logged by Support, but entirely missing from Finance tracking.       |
| Deduction Mismatches | 149           | Amount promised by Support differs from amount disbursed by Finance. |
| Off-Tracker Leakage  | 24            | Escalations with "no ref" originating from unlogged WhatsApp/Email.  |

**Insight:** While a naive net volume calculation (755 Support - 689 Finance) suggests only 66 missing tickets, our deep ID-to-ID database reconciliation reveals the true scale of the failure is **100 dropped tickets**, masked by duplicate records and asynchronous closures.

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### 3. Designing a Zero-Headcount Solution

To permanently bring escalations down, BharatTrip must establish a unified **Single Source of Truth (SSOT)** and automate the communication loop.

#### The Proposed Workflow:

1. **Unified SSOT Tracking:** Migrate both teams onto a single shared database.
  2. **AI-Powered Ingestion:** Deploy an AI agent to monitor the Support Email and WhatsApp. The AI will parse unstructured messages and log them as structured tickets, eliminating "no ref" leakage.
  3. **Automated Communication:** A system trigger automatically emails the travel agent when the status changes to *Paid*, detailing the exact payout amount and any deductions.
  4. **Strategic Trade-offs:** We automate data entry and outbound communication, but deliberately keep *Finance payout execution* and *Support's initial vetting* manual to ensure strict financial compliance.
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### 4. The AI Prototype: Advanced Multi-Agent Architecture

To deliver a production-ready solution, I have developed `app.py`, a zero-friction **Streamlit Web Application** showcasing an advanced Multi-Agent architecture powered by `gemini-3.5-flash`.

[image]

#### Secure Integration (MCP)

In a production environment, the LLM will bridge with the internal SSOT database via the **Model Context Protocol (MCP)**. This guarantees that raw database credentials are never exposed to the LLM's context window, ensuring enterprise-grade security while allowing the AI to read/write records.

#### The Prototype Features

- **Metrics Dashboard:** Live KPIs tracking Escalations and Discrepancies to prove ROI.
  - **HITL Workflow:** A UI for support staff to review AI-generated explanatory messages before they are sent to external agents.
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### 5. Measuring Success

To ensure the solution is working, we will monitor two primary metrics over the next month:

1. **Escalation Rate:** The percentage of refund requests that result in a formal escalation.  
Target: >50% drop within 30 days.
2. **Tracker Discrepancy Count:** The number of tickets marked "Closed" by Support without a matching payout record in Finance. Target: Zero.